

WASP-2b

R-2101

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_111025 · created 2026-08-10T23:28:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455859.6409 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.046
Transit depth (Rp/Rs)^2	2.52 +/- 1.46 [%]
Orbital Inclination (inc)	84.3 +/- 2.75
Airmass coefficient 1 (a1)	0.985 +/- 0.082
Airmass coefficient 2 (a2)	0.028 +/- 0.056
Scatter in the residuals of the lightcurve fit is	8.45 %
Best Comparison Star	#3 - [452, 198]
Optimal Aperture	4.75
Optimal Annulus	10.69
Transit Duration (day)	0.06 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.046 vs 0.1326 (deviation 0.57σ)
Duration fit vs published	0.06 d vs 0.0737 d (deviation 0.39σ)
Mid-time O–C	-4.6 min
Depth SNR	3.5
Residual scatter	8.45 %

Light curve

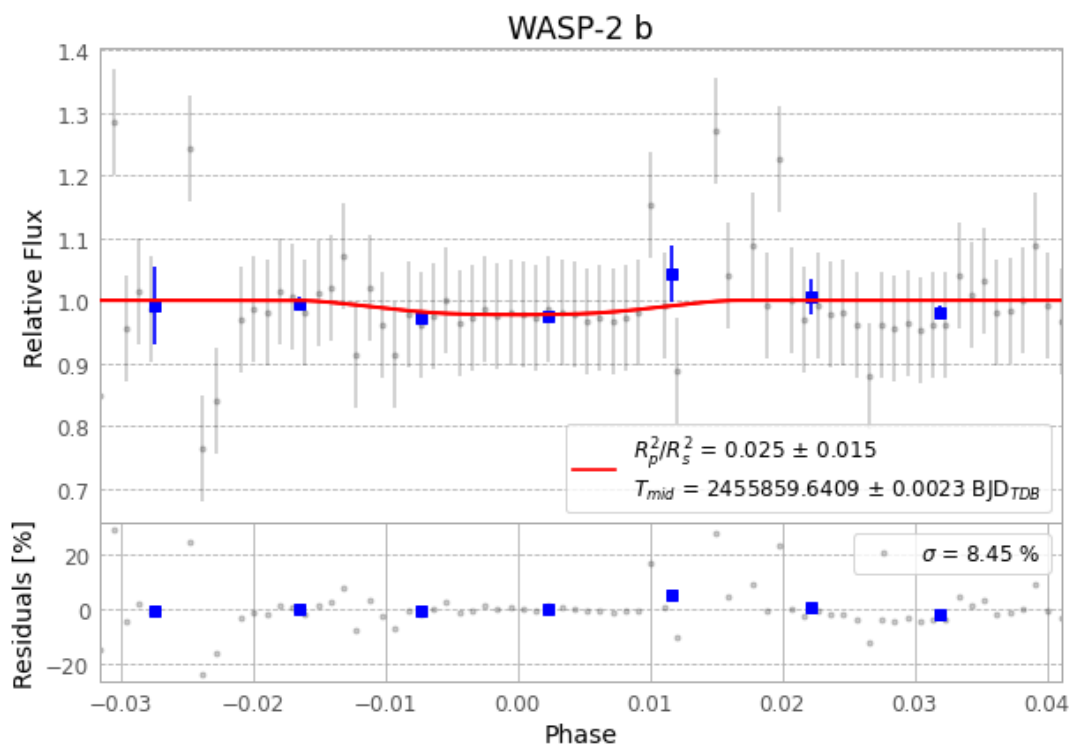


Figure 1 — FinalLightCurve_WASP-2 b_2011-10-24.png (SHA-256 509a0a39f6c46027...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [262, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5f9f5267b6658158...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

76 FITS frames (50 MB), WASP-2111025014212.FITS → WASP-2111025052712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-111025.log (06db10202e5d...), FinalLightCurve_WASP-2 b_2011-10-24.png (509a0a39f6c4...), FinalLightCurve_WASP-2 b_2011-10-24.csv (06c5bc9fedcd...)

Compute resources

Wall time 460.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:28Z.